

iDRAC10: Direct USB Access for Full PowerEdge Server Control

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Summary

Whether you are attempting to configure iDRAC or tackle real-time server troubleshooting, the iDRAC Direct port offers instant access to the iDRAC and server console. No network, no delays, just secure authenticated access. Server administrators can use the iDRAC Direct port—a deceptively simple USB port that augments existing network access—and Quick Sync 2 (using OpenManage Mobile “OMM”) approaches. Far more than just a basic connection, this feature opens a gateway to full advanced out-of-band management, offering a fast and secure method to access, monitor, and maintain PowerEdge servers.

iDRAC Direct empowers you to stay in control with plug-and-play convenience—from data center to remote server closet. The iDRAC Direct USB port, located on the front of every Dell PowerEdge server, enables a direct connection between your laptop and the iDRAC.

This direct-link approach is a secure way to conduct emergency access, or as a replacement to traditional crash cart or KVM switch approaches, especially in environments with restricted network access or strict network protocols security policies. This enables access to the full iDRAC's web UI interface, Remote Access Controller Admin (RACADM) command line, and Redfish RESTful API for server administration.

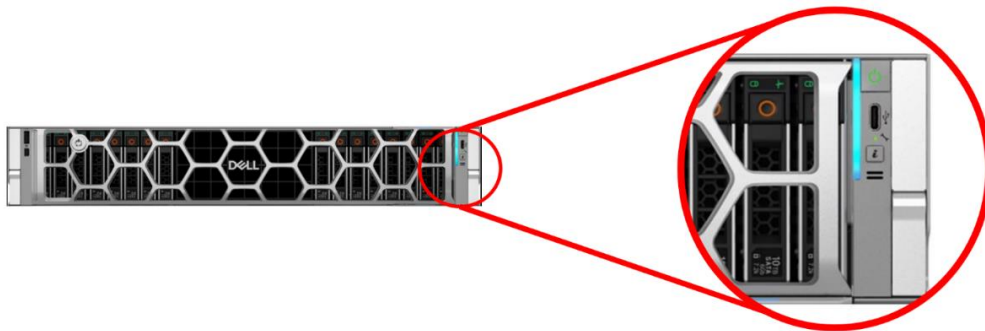


Figure 1. PowerEdge 17th generation server iDRAC direct port location

Note: For PowerEdge 17th generation/iDRAC10 systems, the iDRAC Direct port has been upgraded from the older micro-USB to a USB-C port. To activate the port for management use, press and hold the ⚡ System ID button for five to ten seconds (see below for details).

How It Works

To connect using the Direct iDRAC feature, locate the USB-C port, typically found on the front of the server and marked with a spanner icon. No additional software or license is required to use the iDRAC Direct port.

Requirements

A USB cable with a USB-C interface is required to perform the operation.

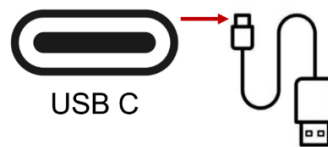


Figure 2. Type of USB cable required for iDRAC direct port use

Steps

1. On your laptop or tablet, turn off all wireless networks and disconnect from any wired network connections.
2. Connect a USB-C cable between the laptop and the iDRAC Direct port on the server.
3. Press and hold the  System ID button (located near the iDRAC Direct port on the front of the server) for five to ten seconds.

Note: This step is new for iDRAC10 on 17th generation PowerEdge servers.



Figure 3. USB-C cable plugged into iDRAC direct port with green LED lit

4. A solid green LED light will appear, indicating that the connection is active and ready for use.
5. Open a web browser on your laptop or tablet and enter the address: 169.254.0.3
Note: This IP address is always the same for iDRAC Direct.
6. When the iDRAC login screen appears, enter your credentials to log in.
Note: If the server was recently shipped from a Dell factory, the default iDRAC password can be found on the rear of the pull-out server information luggage tag. This tag is usually on the front side of the server.

The iDRAC Direct port also supports the automated import of a Server Configuration Profile (SCP) from a USB memory stick. The SCP file can include all or selected settings for iDRAC, BIOS, PERC, and other device firmware configuration such as security settings, boot order, and RAID configuration.

For more details, see the user guide (link provided at the end of this document).

Troubleshooting

If you run into any problems along the way, consider these troubleshooting instructions:

1. Check that the laptop has obtained the IP address 169.254.0.4. Use tools like ipconfig (Windows) or ifconfig (Linux or macOS) to verify.

```
C:\Users\markm>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 5:

    Connection-specific DNS Suffix  . : local
    Link-local IPv6 Address . . . . . : fe80::3813:76a2:433a:7067%15
    IPv4 Address. . . . . : 169.254.0.4
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
```

Figure 4. Designs of the IP address given to the management station by the iDRAC

2. Check if the iDRAC is responding to the ping command from the laptop or tablet over the USB connection:
ping 169.254.0.3
3. Check the iDRAC Direct port configuration. On the iDRAC10 UI, go to:
 - a. **iDRAC Settings > Settings > Management USB settings**
 - b. Ensure the **USB Management Port** is set to **Enabled** (this is the default factory setting).

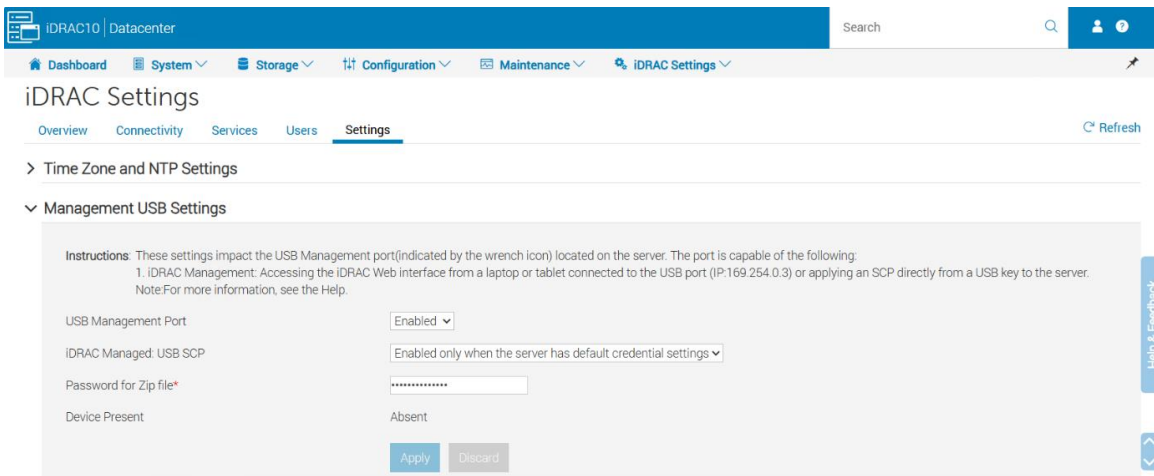


Figure 5. iDRAC10 UI showing iDRAC Direct port configuration settings

4. Check that the laptop or tablet operating system has automatically configured support for the iDRAC Direct port. For example, in Windows 11, the Device Manager should show iDRAC as a **Remote NDIS Compatible Device**, indicating it is ready for use.

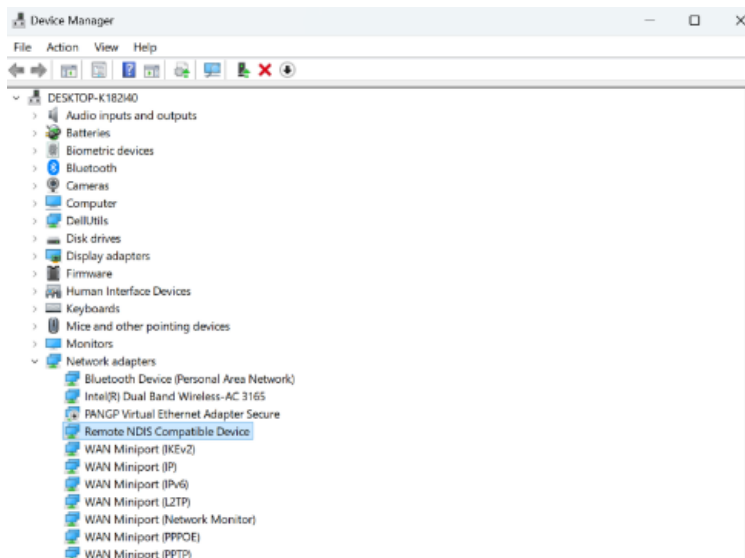


Figure 6. Windows 11 Device Manager showing NDIS enabled

Security in the Spotlight

Dell designed iDRAC Direct with security in mind. The USB connection supports role-based access control and audit logging, ensuring that only authorized personnel can perform management tasks. Additionally, if the iDRAC Direct port is not needed, its functionality can be disabled through the iDRAC UI, CLI, or API.

Conclusion

iDRAC is the go-to management tool for one-to-one management of Dell PowerEdge servers, and the compact iDRAC Direct port unlocks powerful capabilities for configuration and real-time troubleshooting. With just a laptop and a USB-C cable, server administrators can access critical server settings, power controls, and the full remote KVM console—even when the system is offline.

This direct-link method offers a secure way to perform emergency maintenance and management, especially in environments with limited or no network access. This management method serves as a replacement for the legacy LCD bezel (phased out in 17G) and a streamlined alternative to traditional crash carts or KVM switch approaches.

References

- [iDRAC10 User Guide](#)
- [Developer's Dell Technologies Interactive Portal iDRAC10 Redfish API reference](#)
- [iDRAC10 Security Configuration Guide](#)
- [iDRAC10 RACADM CLI Guide](#)
- [iDRAC10 Documentation Collection](#)
- [OpenManage Portfolio Licensing Guide including iDRAC10](#)
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